Detection of Harmful and Jailbreak-Style Prompts Using Transformer-based Classification Models

Spring 2026 DATASCI 207 Section 5

Team members: David Stroud, Floriane Leynaud, Rachel Avram, Trevor Chan

GitHub Repository Link: https://github.com/strouddm/project-jail-break

Email: stroudd@berkeley.edu, floriane_leynaud@berkeley.edu, rachel_avram@berkeley.edu, trevor.chan@berkeley.edu

Abstract

Lightweight classifiers screening user input are the first line of defense against LLM misuse. Many are trained on single-turn data, while real misuse attempts are increasingly distributing harmful content across multiple conversational turns. We compare three architectures: TF-IDF with logistic regression, Qwen3 embeddings with a feedforward neural network, and a fine-tuned ModernBERT transformer. Each was trained on single-turn conversations from WildGuardMix and evaluated on both a held-out single-turn test set and mulit-turn conversations from LMSYS-Chat-1M. On single-turn data, ModernBERT achieved the highest PR-AUC (90.7%), slightly ahead of the embedding-based FFNN (90.3%). When tested on multi-turn conversations both the FFNN and ModernBERT showed improvement in their PR-AUC's, to 92.6% and 91.4% respectively. Due to stronger recall of harmful content in the test set (80% vs 72%), we select the FFNN model as our best performer. We also find evidence that models trained on shorter conversations can generalize well to longer conversations.

Introduction

As large language models (LLMs) become increasingly deployed across search, education, and productivity tools, lightweight classifiers that screen user input are becoming the standard first line of defense against misuse. The stakes are real. In June 2026, a publicized jailbreak bypassed the safeguards of Anthropic's newly released Fable (Mythos class) model, which was consequently pulled from public access following a U.S. government directive [1]. This project asks how well such a classifier can be built, comparing classical text-feature models against a fine-tuned transformer on the task of detecting harmful prompts and jailbreak attempts. But harmful intent is not always confined to a single prompt; a growing class of user jailbreak attempts employ techniques that distribute harmful content across multiple prompts. Each turn in the conversation may look benign in isolation, with only the full context revealing harm. Given that many classifiers evaluate each prompt individually [2], LLMs are particularly vulnerable to such attacks, which may coax them into producing content that, in aggregate, violates safety guidelines [3]. We will first train and test the best classifier possible on single-turn conversation. We are curious to see if there is evidence that classifiers trained on single-prompt conversations generalize well to multi-prompt interactions, or if more specialized training is required when conversations grow longer and more complex. The results can provide valuable insight into best practices for training effective classifiers in the field of LLM safety.

Related work

Developing useful, scalable, and cost effective methods of automated input moderation is a growing field of applied research. Many current techniques rely on single-turn classifiers to flag harm at each step in a conversation. Open tools like Llama Guard [4] and WildGuard [5] were developed specifically to support moderation and determine if a single prompt is harmful. However, in the continuous game of cat-and-mouse, users are finding novel ways around these classifiers by spreading the harm across a conversation. Recent work on multi-turn safety focuses on this by identifying harmful intent spread across a conversation (LMSYS-Chat-1M and SafeDialBench) [6][7]. In essence, no single turn looks harmful, but in aggregate, the moderation policy is bypassed. We use some of the data from this work as the foundation of our project.

Datasets

WildGuardMix [8]: WildGuard's dataset is our primary training resource and establishes a baseline that our work builds upon. WildGuardMix contains synthetic single-turn interactions with LLMs. Before cleaning, it had 88,484 rows and 11 columns, and after cleaning, it had 49,577 rows. The input is the user prompt and LLM response, and the labels are the binary 'prompt_harm_label' and 'response_harm_label' – whether the text is harmful or unharmful. For the purposes of this analysis, we considered a conversation harmful if either the prompt or response was labeled harmful. The class balance was 51.5% and 48.5% harmful-unharmful. We keep the original WildGuardMix train-test split and pull a validation set out of the training data. This yielded train, test, and validation sets of sizes 38,281, 9,571, and 1,725, respectively. To clean the dataset, we removed duplicate prompts. Removing these ensured no prompt appeared in both the train and test sets. We also considered the 'adversarial' (whether or not the prompt is structured like a jailbreak attempt, regardless of harm) and 'subcategory' (type of harm) columns for sub-group analyses, but did not use them in the models.

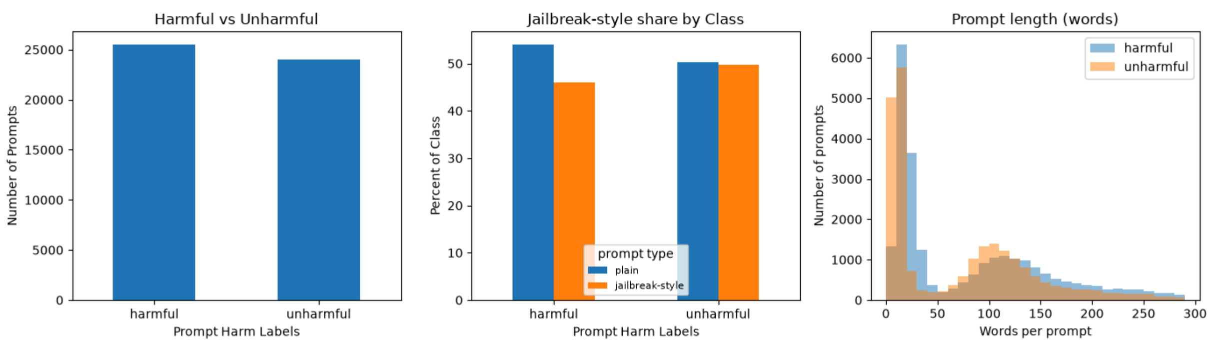


Figure 1. WildGuardMix EDA. (a) class balance; (b) jailbreak-style share within each class; (c) prompt length by class (shown up to 300 words with a few reach ~2,000).

LMSYS-Chat-1M [9]*:* LMSYS contains 1 million rows and 7 columns corresponding to real human conversations with various LLMs across different languages. We used the English conversations for our multi-turn extension. This dataset was used to test how models trained on short interactions performed on unseen longer conversations. The column 'openai_moderation' provided a harm metric for each part of the conversation, including each human prompt and LLM response. Since our analysis was conducted at the conversation level, if any part of the conversation received a harmful rating, we labeled the conversation as harmful. For the multi-turn extension task, we took a sample of 2,000 conversations with a 50-50 harmful-unharmful balance. For the purposes of this analysis, we only considered conversations that had 3 or more turns. This was intended to challenge the models on a dataset that had meaningfully more complexity than the original dataset on which they were trained. This dataset also redacted any names that appeared in the text, using the replacement structure [NAME_x]. To test our models on data that most closely resembles real world text, we filled in these placeholders with randomly selected names from a name dataset [10].

As a final pre-processing step for both datasets, we harmonized conversation structure, placing each conversation in a dict with a {'role': ..., 'content': ...} schema, with role 'user' for prompt and role 'assistant' for LLM response. Then, we converted this structure into string format. One column, designed for the baseline and FFNN models, received further processing (e.g. lowercase, no punctuation), whereas another column, designed for the BERT model, was left unprocessed because BERT benefits from retaining information like punctuation.

Methods

In order to detect harmful content in WildguardMix, we built a baseline model and two improvement models. We used a classification threshold of 0.5. The hyperparameters for our models were chosen on PR-AUC because we care more about our positive class (harmful). For our baseline we used a vectorizer, TF-IDF, and logistic regression. This baseline has the advantage of being quite interpretable, but can't consider context or word order. For our second model, we chose a feedforward neural network (FFNN) powered by embeddings. For the final model, we picked a transformer model that we hoped would capture more of the complexities in the textual data.

Our first model provided a baseline for accuracy, precision, and recall that the other models aimed to beat. The focus in particular was to improve upon recall while keeping accuracy and precision stable.

For our second model, we converted each conversation into vectors. We used Qwen3 due to its large context window (32K). We embedded at the conversation level, which gave us a 1,024-dimension vector per conversation. The 1,024-dimension vector is what the FFNN classifier actually sees in the input layer. Then, the information passes through two hidden layers with relu activation, two dropout layers, and an output layer that uses sigmoid activation to assign each conversation a probability of harm. The embeddings do the heavy lifting on numerically representing over a thousand abstract features across conversations. The FFNN architecture learns which directions in embedding space relate to harmful content. It also learns which combinations of features, which may in isolation be harmless, create harm when they co-occur. This provides a strong advantage over the baseline model when it comes to decoding jailbreak (adversarial) phrasing—better distinguishing between prompts that are harmful and benign regardless of prompt structure.

For our third model, we selected ModernBERT due to its ability to take in a large amount of raw input text and classify it. We leveraged its max token length during training and evaluation stages. A max token length of 1,500 was set for training and evaluation of the single-turn data, covering 99% of the dataset. A max token length of 4,000 was set for evaluation of the multi-turn data, covering 95% of the dataset. It works by taking raw text as input and tokenizing it. The tokens are then converted into a vector of input embeddings which combine token, position, and segment data. These vectors are passed through encoding layers allowing ModernBERT to understand the full contextual meaning.

Experiments, Results and Discussion

<u>Baseline Model:</u> For our TF-IDF and logistic regression we chose to keep stop-words (punctuation was removed during pre-processing) as they are already diluted by TF-IDF, and used an n-gram of (1, 2) to allow a vectorization not only of words, but of pairs of words (such as "ignore previous"). We operated a sweep of two hyperparameters: a pruning parameter, min_df (1, 2, 5), and a regularization parameter, C (4, 8, 16, 32). We selected the best model within a one-standard deviation of the best PR-AUC (0.96) to avoid picking a model that would perform indistinguishably better than the simpler second best one.

Our baseline model had about 128,000 features. It reached an accuracy, macro average recall, and macro average precision of 88% on the validation set. There was a measurable 9-point drop on the single-turn test set: accuracy, recall, and precision fell to 79%. While the validation set is a slice of the training set, the WildGuardMix test set is a smaller, separately drawn partition, so the drop may measure distribution shift rather than memorization. Recall performance was equal (79%) for both classes. For the multi-turn extension test, the model actually performed slightly better, with a precision of 81% and a recall and accuracy of 80%. However, there were large discrepancies between classes, particularly an 11-point decreased recall of harmful content relative to unharmful content (87% vs 74%). While this model shows promise on longer conversations, the class recall gap gives us pause, especially given that catching harmful content is the priority.

The baseline model also missed adversarial conversations more (dropped 11 points in recall), as well as conversations regarding topics where harm is more likely expressed with neutral words (such as copyright violation, stereotypes or private information) rather than with explicit vocabulary – toxic language or defamation related prompts were correctly flagged 95% of the time or more in the test set.

<u>FFNN with Embeddings:</u> We fed the embeddings into the FFNN model for training. After performing hyperparameter tuning based on PR-AUC (0.96), we landed on 128 and 64 nodes for the hidden layers, a 0.5 dropout rate, a 0.001 learning rate, and 15 epochs. This model has 139,521 parameters.

On the validation set, this model achieved macro average precision, recall, and accuracy scores of 90%. There was again a drop in performance on the test set, though relatively small: precision was 86%, recall was 85%, and accuracy was 86%. However, there was a notable gap in performance between the harmful and unharmful classes on recall: 80% vs 90%. This means that the model barely improved upon the baseline in harmful content detection. On the multi-turn dataset, the model achieved an 86% on precision, recall, and accuracy. This represented a small boost in performance compared to the test data from WildGuardMix, which is surprising given that the model was trained on this dataset. The class recall gap shrunk to 6% (89% unharmful vs 83% harmful), which is an improvement relative to the single-turn test results of the FFNN model, as well as the baseline model's multi-turn performance.

Compared to the baseline model, the FFNN model improved on all metrics. However, a 10-point gap remained in class recall performance in the single-turn test case, which is especially concerning given that our priority is to catch harmful content. However the discrepancy in performance from adversarial framing, which was found in the baseline model, disappeared in the FFNN model, suggesting that the embeddings did indeed help the model separate phrasing from harm. The strongest subcategory, "causing material harm by disseminating misinformation," had a recall of 100%. This category would not necessarily have obviously harmful words associated with it, suggesting that the model may be better than the baseline at detecting subtlety. However, some categories still struggled, including "social stereotypes and unfair discrimination," which had a recall of 63%. The fact that the model performed the same or better on the multi-turn dataset across the board, with a smaller recall imbalance for harm, suggests that this model generalizes well to new, more complex data—at least relative to the baseline.

<u>ModernBert Model:</u> ModernBERT was fed the raw training, validation, and test split data. We ran hyperparameter tuning on the model, based on the best PR-AUC score, to determine the best values for learning rate and weight decay, landing at 3.07e-5 and 0.038 respectively. The model attained an accuracy, macro average recall, and macro average precision of 93% on the validation set. There was a mild difference in recall performance for the unharmful (97%) and harmful (90%) classes. Like the baseline model, there was a sizable drop in accuracy and recall when moving from validation to the single-turn test set. On single-turn data, the model had 85% accuracy, 84% recall, and 87% precision. The model had a recall of 96% on unharmful and 72% on harmful. In regards to recall on adversarial vs normal conversations, the model performed similarly across the two, but overall slightly worse when compared to the FFNN model. The model performed very similarly on the multi-turn evaluation set as it did on the single-turn evaluation set, with metrics of 83% for accuracy, 83% for recall, and 85% for precision. Similar to the single-turn evaluation, the multi-turn evaluation also had a recall of 72% for harmful. This model performed worse on recall than the baseline model and FFNN model most likely due

to the fact that it was trained on general domain text, and failed when faced with nuanced or specific jargon.

Table 1. Results on Single-Turn Test Set

Model	Accuracy	Recall (harmful)	Precision	PR-AUC
TF-IDF + LogReg (min_df =2 and C= 8)	79%	79%	79%	85%
Qwen3 + FFNN (5 layers)	86%	80%	86%	90.3%
ModernBERT (22 layers)	85%	72%	87%	90.7%

Conclusion

The FFNN model with textual embeddings had the best performance on accuracy and harmful recall, while the ModernBERT model has the best precision and PR-AUC, but only by a very small margin (1% and 0.4%, respectively). However, the BERT model's recall performance suffered notably on harmful content. Given that we prioritize finding harmful content, we select the FFNN model as our strongest performer. Even for the FFNN model, there remain concerns, including a 10-point gap in recall on harmful vs unharmful content. In future iterations of this project, we would be interested in lowering the classification threshold for deeming a conversation harmful to see if this helps address the issue. We could also train the model on a larger pool of data from new sources to augment its harm detection.

Our project also provides evidence that models trained on single-turn conversations can generalize performance to longer, more complex conversations. This suggests that classifiers trained on smaller datasets can be applicable to new datasets without the need to start from scratch, saving time, money, and compute. While there is still much room for improvement in our results, we feel that this project can act as a starting point for more research into best practices for training generalizable harm classifiers.

Contributions

Flo contributed to this project by coming up with the idea, provided notes and research on the topic of jailbreaking that helped the team confirm we wanted and could do it. She found the LMSYS dataset used for multi-turn extension and conducted EDA on both dataset. Contributed to the experimental design, then to building and running the baseline model and to the error analysis.

Rachel contributed to this project by writing the preprocessing pipeline for the two datasets, as well as an EDA file for the train/test sets. She also created the FFNN model notebook. She helped polish the GitHub repo, as well as contributed to the final presentation and written report.

Trevor contributed to this project conducting EDA on the SafeDial dataset. He built and ran the ModernBERT model portion of the experiment, performing analysis on the results. He also contributed to portions of the final report and presentation.

David contributed to this project by setting up the repository, creating the initial README docs, providing input to the CONTRIBUTING docs. Created the script to aid database download for onboarding. Created the EDA for WildGuardMix to set up the team for pre-processing work. Worked on the embedding model selection and created the embeddings for the FFNN model. Contributed to portions of the final report and helped clean up the repo.

References:

[1] Anthropic. "Claude Mythos Preview Access." Anthropic, 2025.
https://www.anthropic.com/news/fable-mythos-access

[2] Tamma, Rohit. "The Attack That Jailbroke GPT-5 and What It Means." Substack, 2025.
https://rohittamma.substack.com/p/the-attack-that-jailbroke-gpt-5-and

[3] SpecterOps. "This One Weird Trick: Multi-Prompt LLM Jailbreaks — Safeguards Hate It." SpecterOps Blog, 5 Sept. 2025.
https://specterops.io/blog/2025/09/05/this-one-weird-trick-multi-prompt-llm-jailbreaks-safeguards-hate-it

[4] Inan, Hakan, et al. "Llama Guard: LLM-Based Input-Output Safeguard for Human-AI Conversations." arXiv preprint arXiv:2312.06674 (2023).

[5] Han, Seungju, et al. "WildGuard: Open One-Stop Moderation Tools for Safety Risks, Jailbreaks, and Refusals in LLMs." arXiv preprint arXiv:2406.18495 (2024).

[6] Zheng, Lianmin, et al. "LMSYS-Chat-1M: A Large-Scale Real-World LLM Conversation Dataset." arXiv preprint arXiv:2309.11998 (2023).

[7] Ran, Hongye, et al. "SafeDialBench: A Fine-Grained Safety Evaluation Benchmark for Large Language Models in Multi-Turn Dialogues with Diverse Jailbreak Attacks." arXiv preprint arXiv:2502.11090 (2025).

[8] Lo, Bill Yuchen, et al. "WildGuardMix." Hugging Face Datasets, Allen Institute for AI, 2024.
https://huggingface.co/datasets/allenai/wildguardmix

[9] Zheng, Lianmin, et al. "LMSYS-Chat-1M." Hugging Face Datasets, 2023.
https://huggingface.co/datasets/lmsys/lmsys-chat-1m

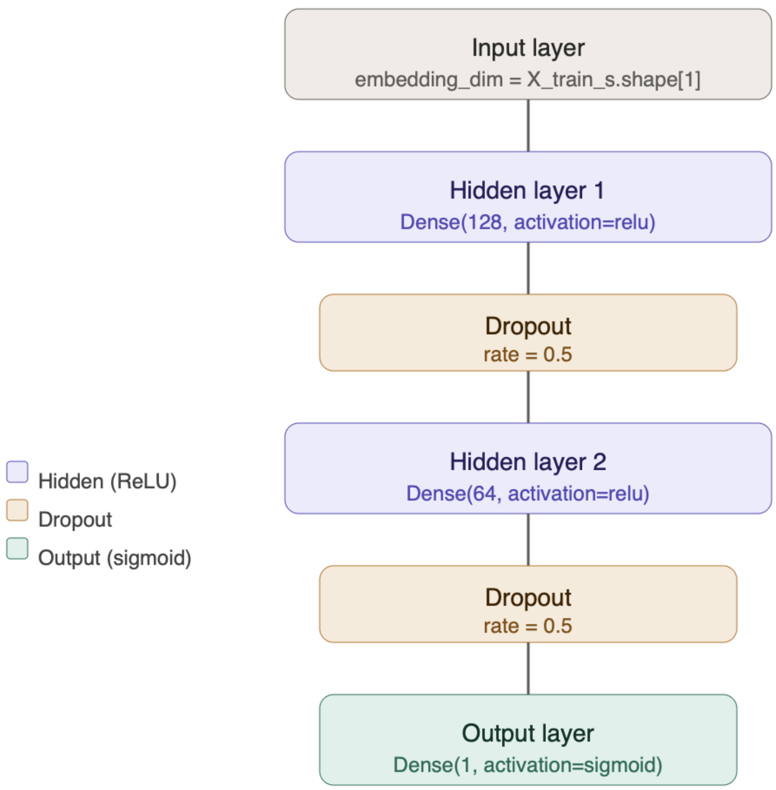
[10] Kaggle. "United States Baby Names 1880–2017." Kaggle Datasets, 2017.
https://www.kaggle.com/datasets/kaggle/us-baby-names

Appendix:

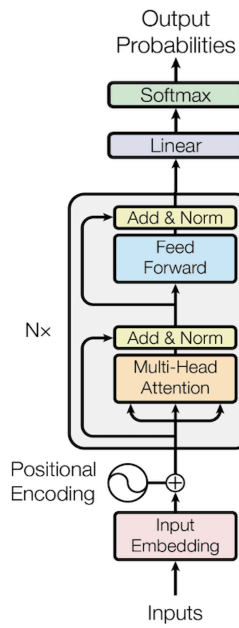
Table 1. Experimental Design

Architecture	Single-turn training	Single-turn test	Multi-turn test
TF-IDF + Logistic Regression	38,281 / 9,571 Validation / Test	1,725 Wildguard Test	2,000 LMSYS test
Qwen3 Embeddings + FFNN	38,281 / 9,571	1,725 Wildguard Test	2,000 LMSYS test
ModernBERT (Transformer)	38,281 / 9,571	1,725 Wildguard Test	2,000 LMSYS test

FFNN Model Architecture



ModernBERT Architecture



Recall: Adversarial vs Normal (All 3 Models)